

2121 Tumor Bed Dynamics after Surgical Resection for Brain Metastases: Implications for the Timing of Postoperative Radiosurgery

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Purpose/Objective(s): Stereotactic radiosurgery (SRS) directed to the tumor bed after surgical resection of a brain metastasis decreases local tumor recurrence and delays whole brain radiotherapy in appropriately selected patients. SRS is commonly delivered 2-4 weeks after surgery, but optimal timing is not known. This study characterizes resection cavity dynamics and determines the incidence of tumor progression in the interval between surgery and SRS.

Materials/Methods: For each patient we obtained: (1.) preoperative MRI, (2.) MRI scan within 24 hours after surgery, and (3.) MRI scan for radiosurgery planning (within 1 week before SRS). MRI scans included multisequence, multiplanar views of the brain before and after administration of gadolinium. Scans were reviewed by a neuroradiologist. Tumor volumes were determined by contouring the contrast enhancing lesions on axial slices of the T1 post-contrast MRI obtained prior to surgery. Resection cavity volumes were determined by contouring the surgical defect and any associated contrast enhancement on the T1 post-contrast axial MRI from the postoperative and the radiosurgery planning MRI scans. Contouring was carried out on Stentor iSite 3.3.2 software using the freehand ROI tool. The surgical resection was categorized as a gross total or subtotal resection (GTR vs. STR) based on the 24 hour postoperative MRI. Tumor progression prior to SRS was determined by comparing the postoperative MRI to the SRS planning scan.

Results: Forty-three surgical cavities were analyzed (2 patients had SRS to 2 cavities). The interval between the postoperative MRI and radiotherapy planning MRI averaged 23.9 days. SRS was delivered (mean) 29.8 days after surgery. 37 resections were GTR and 6 were STR. The mean tumor volume prior to surgery was 14.23 cm³. The mean resection cavity volume in the immediate postoperative period was 8.53 cm³ and at the time of SRS planning was 8.77cm³. In the interval between surgery and SRS, 20 cavities (46.5%) were stable in size as defined as a change of <2 cm³, 10 cavities (23.3%) collapsed by >2 cm³, and 13 cavities (30.2%) enlarged by >2 cm³. In the interval between surgery and SRS, 9 patients had local tumor progression, 4 had progression elsewhere in the brain, 1 had local and elsewhere progression and 29 had stable disease.

Conclusions: In the weeks following resection, surgical cavities are dynamic, changing in size in the majority of patients (53%). These observations support obtaining MRI imaging for radiosurgery planning close to the time of SRS. In addition, local progression prior to SRS arises in some patients (23%) prior to SRS, suggesting that delaying SRS risks tumor progression without the benefit of cavity collapse.

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2122 A Flexible Multimodal Approach to Eye Melanoma: Patterns of Care and Related Complications

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Purpose/Objective(s): to evaluate the efficacy and safety of a flexible radiotherapy approach for uveal malignant melanomas (UMM).

Materials/Methods: a retrospective analysis was performed on a series of 100 consecutive patients treated between 2003 and 2010 in a single institution. Main outcome measures were survival rate, enucleation rate, local tumor control, visual outcomes and complications, scored according to the Common Terminology Criteria for Adverse Events v4.0 (CTCAE).

The treatment was planned depending on tumor size and location, using a range of multimodal options such as brachytherapy plaque (Ruthenium 106 and/or Iodine 125), stereotactic irradiation (SRS), trans-pupillary thermo-therapy and intravitreal administration of anti-VEGF agents.

Brachytherapy was the treatment of choice in small- (less than 3mm) and medium-sized (between 3 and 10 mm) tumors. A 106-Ru plaque was used for T1-T2 lesions with a single administration with a prescription dose of 100-120 Gy to tumor apex. For medium-sized lesions a single fraction with a 125-I plaque was used with a prescription dose of 85 Gy to COMS point for lesions less than 5 mm and to tumor apex for larger lesions. In selected cases, two subsequent fractions with 106-Ru and a 125-I plaque were used to keep the scleral dose with acceptable toxicity levels while ensuring an optimal dose coverage. In the presence of large (greater than 10 mm) the treatment of choice was SRS with a dose of 30 Gy in a single fraction. A sequential brachytherapy treatment was performed 6 weeks after SRS with a prescription dose of 45 Gy to COMS point.

Results: One hundred consecutive patients were managed with conservative therapy and constitute the basis of the present analysis. According to the Co-operative Ocular Melanoma Study (COMS) classification, at baseline, small-, medium- and large-sized tumors were 32%, 64%, 10%, respectively. The median follow-up was 35.4 months. The 5-year overall survival was 95.3%. The local recurrence rate was 3.7%. Vision preservation was achieved in 84% of cases. Treatment-related major complications occurred in 6% of cases (2 severe glaucomas, 1 scleral necrosis, 2 radiation retinopathy, 1 optic neuropathy). Enucleation was performed in 4% of cases due to uncontrolled neovascular glaucoma (2 cases) or local recurrence (2 patients).

Conclusions: The flexible approach to uveal melanomas utilized in our institution achieves a high local tumor control rate with a low complication rate by reducing the over-irradiation of healthy tissues with optimal quality of life outcomes. The survival rates appear similar to previously published results.

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2123 Presence of Hemosiderin Ring on T2-weighted MRI for Renal Cell Carcinoma Brain Metastases Prior to Radiosurgery is Associated with Higher Risk of Post-SRS Radionecrosis

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